

Tinker Tailor LLM Spy: Reconstructing "Deleted" Chats & Hijacking Sessions from Chromium LevelDB Caches

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I. The Ephemeral AI Fallacy

- **The Mirage of Deletion:** User interfaces suggest that clicking 'Delete Chat' permanently wipes data. Cloud-side pointers are removed, but raw local disk artifacts remain.
- **Client-Side Persistence:** Web browsers (Chrome, Edge) cache conversation logs inside unencrypted IndexedDB LevelDB SSTables and log segments.
- **The Threat Vector:** Low-privilege local processes or malware can silently scrape unsubmitted keystroke inputs and active session keys without admin elevation.

II. Reversing LevelDB & V8 Storage Schemas

- **LevelDB LSM-Tree Persistence:** LevelDB relies on an append-only Write-Ahead Log (.log) and uncompactd SSTables (.sst/.ldb). Deleted data stays intact in unmerged tables.
- **V8 ValueSerializer Format:** Dissecting V8 string tags (OneByteString vs TwoByteString UTF-16LE anomalies) and nested object boundaries.
- **Credential Exposure:** Active authorization session tokens (JWTs) are stored in plaintext in IndexedDB LevelDB caches rather than secure OS keychains, enabling session hijacking.

III. Sub-50ms Infostealer Vector (verity_stealer.py)

- **High-Speed Headless Execution:** A zero-dependency Python script (verity_stealer.py) harvests local LLM caches in-memory without starting heavy browser drivers.
- **41 Millisecond Latency:** Complete extraction of 485+ prompts and session cookies in 41ms.
- **EDR Evasion:** Bypasses Windows file locks in read-only mode and completes execution before standard EDR disk I/O monitoring triggers alerts.

IV. Live Verity Dashboard & DLP Engine

- **Multi-Platform Support:** Verity unifies local telemetry carving across 7 platforms: ChatGPT, Claude AI, Gemini, DeepSeek, Perplexity, Cursor IDE, and VS Code Copilot.
- **Unsubmitted Claude TipTap Keystrokes:** Captures text entered in the web UI prompt buffer before submission.
- **Shadow AI DLP Alerts:** Integrated regex-based scanner parsing carved chats on-the-fly, flagging leaked AWS keys, slack tokens, and PII inside chat bubbles.

V. Conclusion & Defensive Mitigations

- **Enforce Encryption:** Apply OS-level cryptographic protections (like DPAPI or Keychain) to browser IndexedDB LevelDB paths.
- **Block Wiping:** Mandate explicit zeroization of LevelDB log blocks upon chat deletion.
- **Open-Source Tool Release:** The complete Verity framework is available on GitHub:
<https://github.com/prajwalchowdary2/tinker-tailor-llm-spy>